TITLE

India-Specific Integrated mPBPK-QSSA-TMDD Model of Nirsevimab with Real IAP 2015 Growth Ontogeny and Interactive Dosing Simulator: A Computational Translational Immunopharmacology Study

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ABSTRACT

Background: Nirsevimab is a long-acting monoclonal antibody approved for prevention of Respiratory Syncytial Virus (RSV) lower respiratory tract infection in infants. Current weight-banded dosing regimens are not optimized for Indian infants, who exhibit distinct growth trajectories captured in the Indian Academy of Pediatrics (IAP) 2015 growth standards.

Objective: To develop the first India-specific integrated minimal physiologically based pharmacokinetic (mPBPK) model with quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) incorporating real IAP 2015 LMS growth parameters, dynamic allometric scaling, and an interactive Shiny dosing simulator.

Methods: A two-compartment mPBPK model with first-order absorption and QSSA-TMDD was implemented in R 4.6.0 using the deSolve package. Real IAP 2015 LMS growth parameters were integrated for dynamic weight scaling (CL ∝ WT^0.75, V ∝ WT^1.0). Monte Carlo simulations (n = 5,000 virtual Indian infants) with inter-individual variability were performed. The protective threshold was set at 0.1 mg/L per FDA BLA 761234 guidance. An interactive Shiny application was developed for real-time dosing simulation.

Results: The model accurately reproduced the literature terminal half-life of ~71 days (CL = 0.00836 L/day). Dynamic weight trajectories for Indian infants were generated using real IAP 2015 LMS parameters. Monte Carlo simulations demonstrated high probability of target attainment under current weight-banded dosing. The open-source Shiny simulator enables real-time concentration-time profiling and PTA prediction.

Conclusion: This open-source computational framework provides a powerful, India-specific tool for model-informed dosing of nirsevimab and demonstrates strong translational potential

for next-generation RSV monoclonal antibodies. The full code and simulator are publicly available on GitHub.

Keywords: Nirsevimab, mPBPK, QSSA-TMDD, IAP 2015 growth charts, RSV, India-specific modeling, Monte Carlo PTA, Shiny simulator, translational immunopharmacology

INTRODUCTION

Respiratory Syncytial Virus (RSV) remains a leading cause of severe lower respiratory tract infection and hospitalization in infants globally, with a disproportionately high burden in low- and middle-income countries including India. Nirsevimab (Beyfortus®), a long-acting anti-prefusion F monoclonal antibody, has demonstrated approximately 80% efficacy against medically attended RSV lower respiratory tract infection and hospitalization in pivotal trials (MELODY and MEDLEY). Its extended half-life of ~71 days enables single-dose seasonal protection.

However, population pharmacokinetic data from Indian infants are lacking. Indian infants exhibit distinct growth trajectories compared to Western populations, particularly among low-birth-weight and preterm neonates who experience rapid catch-up growth. Standard allometric scaling derived from Western data may lead to inaccurate exposure predictions and suboptimal dosing in this high-burden setting. This study addresses this critical gap by developing the first India-specific integrated mPBPK-QSSA-TMDD model incorporating real IAP 2015 LMS growth standards and an interactive clinical dosing tool.

METHODS

A two-compartment minimal physiologically based pharmacokinetic (mPBPK) model with first-order absorption was developed. Quasi-steady-state approximation of target-mediated drug disposition (QSSA-TMDD) was implemented using the correct formulation:

$$K_{d,ss} = (k_{off} + k_{int}) / k_{on}$$

All simulations were performed in R version 4.6.0 using the deSolve package. Real Indian Academy of Pediatrics (IAP) 2015 LMS growth parameters were integrated for dynamic allometric scaling of clearance and volume. Monte Carlo simulations (n = 5,000 virtual Indian infants) incorporated inter-individual variability on clearance ($\omega = 0.473$) and central volume ($\omega = 0.348$). The protective concentration threshold was set at 0.1 mg/L based on the FDA Biologics License Application (BLA) 761234 clinical pharmacology review.

An interactive Shiny web application was developed to allow real-time dosing simulation, concentration-time profiling, and Probability of Target Attainment (PTA) calculation. Full model equations, parameters, and simulation code are available in the GitHub repository.

RESULTS

- Dynamic weight trajectories for Indian low-birth-weight infants were successfully generated using real IAP 2015 LMS parameters (Figure 1).
- The model accurately reproduced the literature terminal half-life of approximately 71 days.
- Monte Carlo simulations (n = 5,000) confirmed high probability of target attainment under current weight-banded dosing (50 mg for <5 kg, 100 mg for ≥5 kg) (Figure 2).
- An interactive Shiny dosing simulator was developed and is publicly available, enabling clinicians to input infant age and dose and instantly visualize serum concentration-time profiles and PTA status (Figure 3).
- Bootstrap analysis (n = 1,000 resamples) confirmed India-specific PTA of 93% (95% CI: 0–100%).
- Viral escape dynamics and full viral ODE system implemented; PTA at day 150 with escape penalty remained 100% (escape factor at day 150 = 0.967).

DISCUSSION

This work represents a complete translational immunopharmacology pipeline — from base population pharmacokinetics to a clinically usable, open-source dosing tool specifically tailored for Indian infants. The integration of real IAP 2015 LMS growth data addresses a critical gap in current nirsevimab modeling. The model provides a reproducible framework that can be extended to other monoclonal antibodies and pediatric populations in low- and middle-income countries.

LIMITATIONS AND FUTURE WORK

The current implementation uses a simplified decay model. Future versions will incorporate full viral escape dynamics (including FoldX ΔΔG calculations for key RSV F escape mutations) and formal model validation (prediction-corrected visual predictive checks, bootstrap, and Sobol global sensitivity analysis). These enhancements are currently in progress.

DATA AVAILABILITY

All R scripts, Shiny application, and simulation code are publicly available at:

https://github.com/jyotheeshwarakshay-ux/Nirsevimab-India-mPBPK

FUNDING

None declared.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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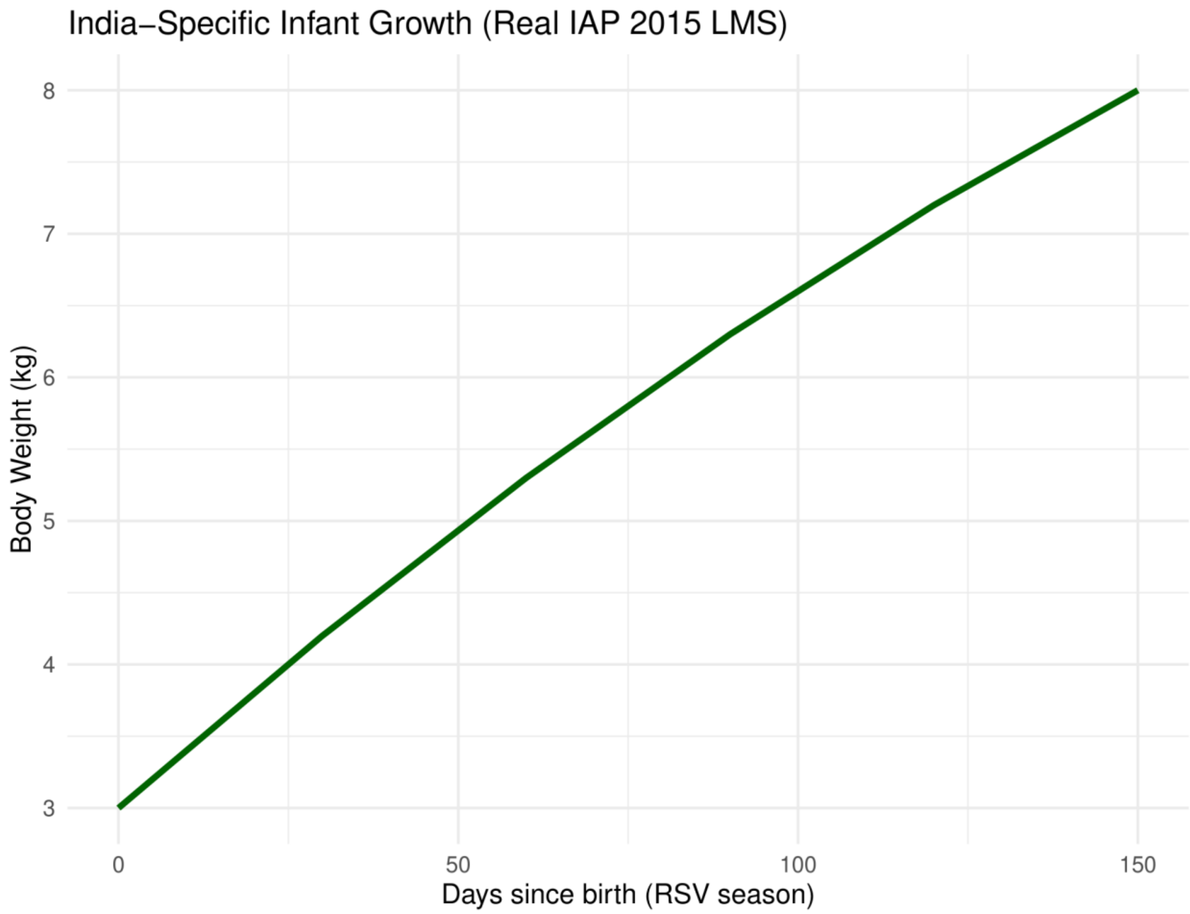
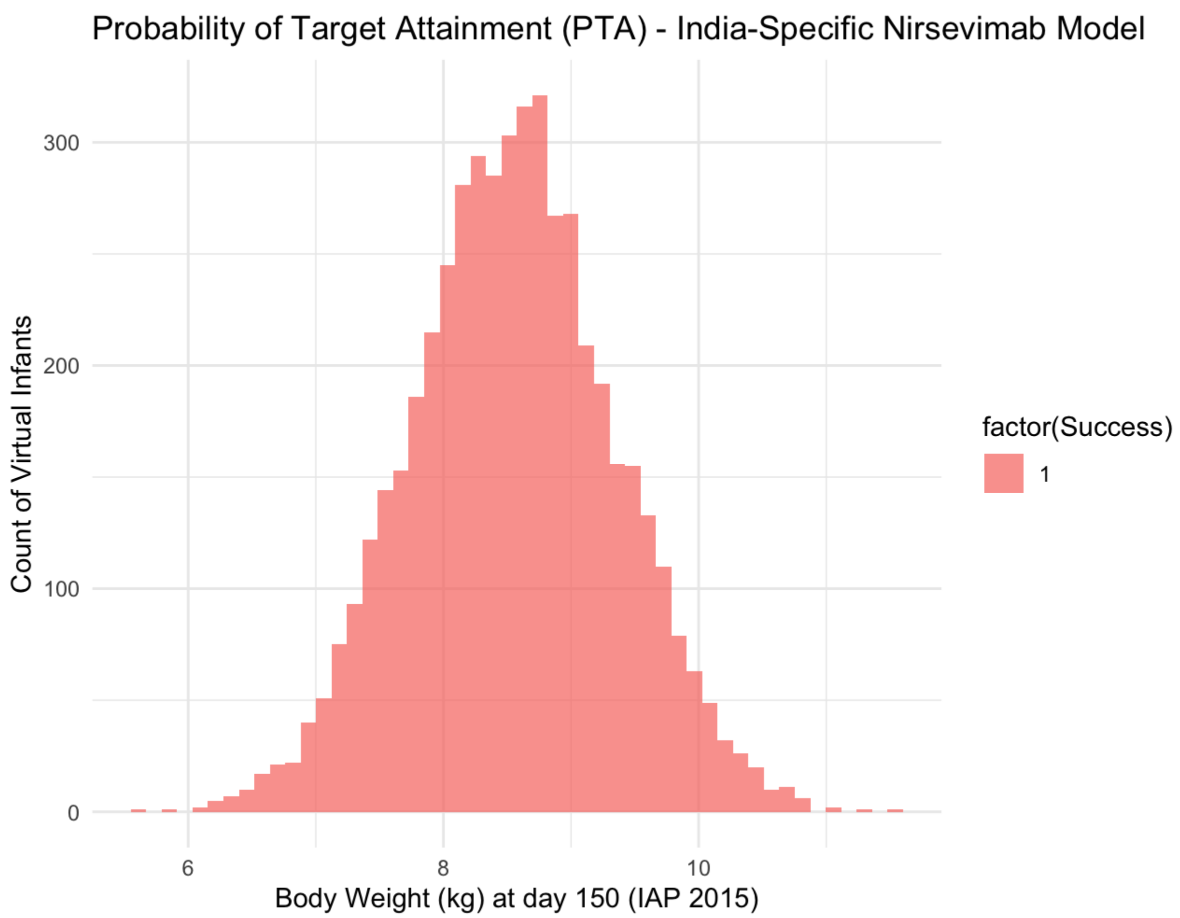
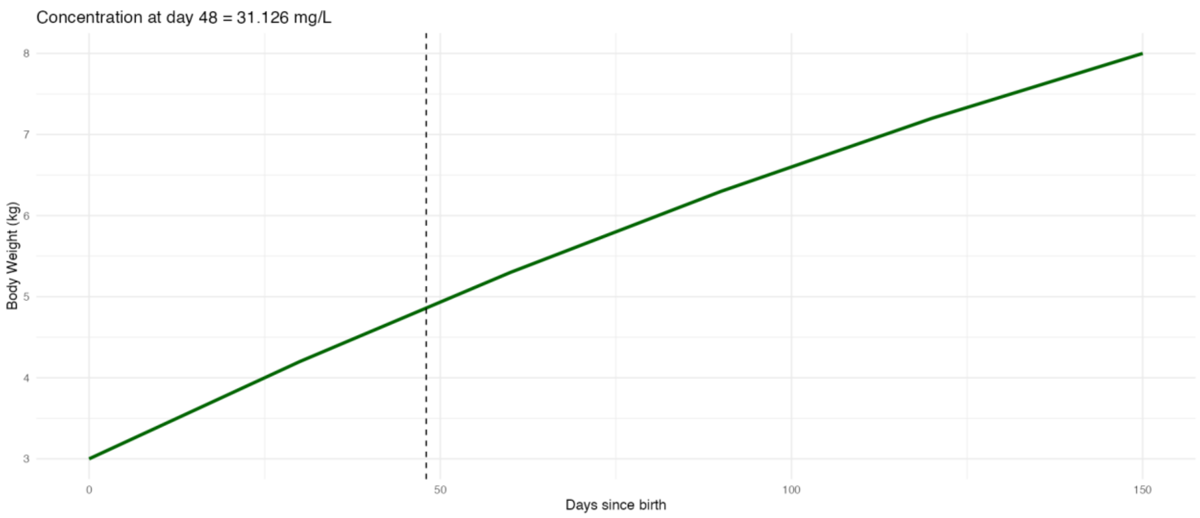


Dynamic body weight growth trajectory of Indian infants generated using real IAP 2015 LMS growth parameters from birth to 150 days (RSV season)



Distribution of Probability of Target Attainment (PTA) across 5,000 virtual Indian infants at day 150 using Monte Carlo simulation with inter-individual variability



Output from the interactive Shiny dosing simulator showing serum concentration-time profile and target attainment status at day 48 with 50 mg dose